

Jannatul Ferdous

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Professional Summary

Experimental quantum photonics researcher and Ph.D. candidate with 6+ years of experience designing, building, and automating optical and photonic experiments for quantum sensing, imaging, and precision measurement. Experienced in optical system design, photon-number-resolving detection, plasmonics, experimental hardware prototyping, multiphysics simulation, LabVIEW-based automation, and Python-based data analysis. First and co-first-author of peer-reviewed research published in *Nature*, *ACS Photonics*, and *Physical Review Applied*. Peer reviewer for journals in quantum optics and photonics.

Technical Skills

Quantum Optical Techniques	Photon-number-resolving detection, optical characterization of materials, quantum state characterization, quantum sensing, quantum imaging, quantum metrology
Optical Systems	Interferometry, advanced optical system design, free-space optics, confocal microscopy
Detectors & Instrumentation	SNSPD, APD, SLM, DMD, lasers, motorized stages, optical components, data acquisition systems, time tagger
Photonics	Plasmonics, Nanophotonics, photonic waveguides, LSPR/SPR
Computational Modeling	COMSOL Multiphysics, ANSYS Lumerical FDTD
Programming & Technical Computing	Python, MATLAB, Mathematica, LabVIEW, LaTeX
Design & Fabrication	Blender 3D modeling, 3D printing, experimental hardware prototyping, FIB fabrication

Professional Experience

Graduate Research Assistant — Ph.D. Research

2021–Present

Louisiana State University, Quantum Photonics Group

Project 1 — Quantum Statistical Plasmonic Metacrystals

Designed and experimentally characterized room-temperature plasmonic metacrystals to engineer statistical bands and control the quantum statistics of multiparticle light, using ANSYS Lumerical FDTD, PNRD data processing, quantum-state characterization, SNSPDs, motorized stages and confocal microscopy imaging. *Nature* (2026) — “Quantum Statistical Plasmonic Metacrystals”, Co-first Author.

Project 2 — Quantum-Plasmonic Hydrogen Sensing

Led experimental development and system integration of a multiparticle quantum-plasmonic hydrogen sensor, including hydrogen sensor calibration, synchronized data acquisition and control using LabVIEW, PNRD data processing, confocal microscopy imaging and optical characterization, and Fisher information analysis, achieving $\sim 100\times$ faster response and $\sim 46\times$ improved detection limit over the classical protocol. *Manuscript under review, Nature Communications*, “Scalable Quantum Sensing with Robust Multiparticle Plasmonic Systems”, (2026), First Author.

Project 3 — Quantum Reservoir Computing on a Plasmonic Platform

Leading an ongoing experimental development of a plasmonic quantum reservoir for multidimensional information processing, using OAM and polarization control, SLM-based spatial-mode manipulation, PNR detection with APDs, time-tagged photon measurements, and optical correlation analysis.

Project 4 — Evolution of Multiphoton Quantum Coherence

Led experimental investigation of multiphoton quantum coherence during free-space propagation, combining SLM-based polarization control, quantum-state tomography, photon-number-resolving detection, and theoretical analysis using MATLAB and Mathematica. *Published in ACS Photonics* “Emergence of Multiphoton Quantum Coherence by Light Propagation”, (2024), First Author.

Project 5 — Multiphoton Quantum Imaging Using Natural Light

Contributed to experimental development and data collection, including raster scanning, for a multiphoton quantum imaging platform using thermal light, photon-number-resolving detection, and compressive sensing to reconstruct high-contrast images under strong background noise. *Published in Applied Physics Reviews* (2025) — “Multiphoton Quantum Imaging Using Natural Light.”

Additional Research Projects

Contributed to additional projects in plasmonic biosensing, quantum phase measurement using homodyne interferometry, photonic-waveguide information processing, and pseudo-thermal light-source development, applying optical system design, interferometry, detector instrumentation, quantum-state characterization, and experimental troubleshooting.

Graduate Research Assistant — M.S. Research [2019—2021]
University of Texas Rio Grande Valley

Project 1 — Tunable Non-Hermitian Acoustic Filter
Designed, fabricated, and experimentally characterized a tunable non-Hermitian acoustic filter, including 3D-printed waveguide development, acoustic measurements, and COMSOL simulation validation. *Published in Physical Review Applied* “Tunable Non-Hermitian Acoustic Filter”, (2021), Co-first Author.

Project 2 — Robust Zero-Energy Extended States
Theoretically and experimentally demonstrated robust zero-energy bulk states in a non-Hermitian acoustic system, using COMSOL modeling, acoustic waveguide characterization, controlled disorder, frequency-domain measurements, and transmission-spectrum analysis. *Published in Physical Review Applied* “Observation of Robust Zero-Energy Extended States”, (2023), First Author.

Graduate Teaching Assistant
Louisiana State University 2021–2023

- Instructed undergraduate physics laboratories and mentored students in experimental techniques, instrumentation, and quantitative data analysis.
- Supported undergraduate course instruction, including lectures and coursework.

University of Texas Rio Grande Valley 2020–2021

- Delivered online laboratory instruction and guided students through simulation-based experiments and data analysis.
- Developed lectures, laboratory assignments, and instructional materials.

Education

Ph.D. Physics (Quantum Photonics)	Louisiana State University (Expected 2027)
M.S. Physics	University of Texas Rio Grande Valley
M.S. Physics	University of Dhaka
B.S. Physics	University of Dhaka

Conferences and Presentations

Nonclassical Multiphoton Plasmonic Sensing — IEEE RAPID (2025), Oral Presentation
Observation of the van Cittert-Zernike Modification of Quantum Coherence — LAOP (2024), Oral Presentation
Quantum Photonic Technologies based on Natural Lightme — LSU Institute for Energy Innovation(talk) (2024), Oral Presentation.
Inducing Quantum Coherence by Light Propagation — LSU Graduate Research Conference (2023), Oral Presentation.

Leadership & Awards

Media Coordinator 2025–2026
Bangladesh Students Association, Louisiana State University

International Student Liaison 2022–2023
Department of Physics, Louisiana State University

- Represented international students and coordinated communication with faculty and departmental leadership.
- Coordinated media communications, event promotion, and organizational outreach.

Presidential Graduate Research Assistantship Award 2019
University of Texas Rio Grande Valley